

Credit Card Payment System

RBAC, Fraud Detection, Analytics & Monitoring — Feature Report

Name: Aprajit Roshan

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Task: Implement role-based access control, fraud detection and alerting, card usage analytics, and system monitoring with export in the CreditPay Credit Card Payment System

This report documents the security, fraud, analytics and monitoring features added to the CreditPay Credit Card Payment System (Django REST Framework backend, FastAPI payment service and a web frontend).

Role-Based Access Control with three roles — **Admin**, **Support** and **Read-Only** — now protects the card management, transaction and analytics APIs, and admin actions such as card block/unblock and credit-limit updates are written to an **audit log**. A rule-based **fraud detection** service flags suspicious payments, stores them in a fraud log and emails the customer. The **Spending Analytics** page offers monthly, category-wise and credit-utilization charts with PDF/CSV export, and the admin dashboard shows live **system health** metrics.

User Roles & Permissions

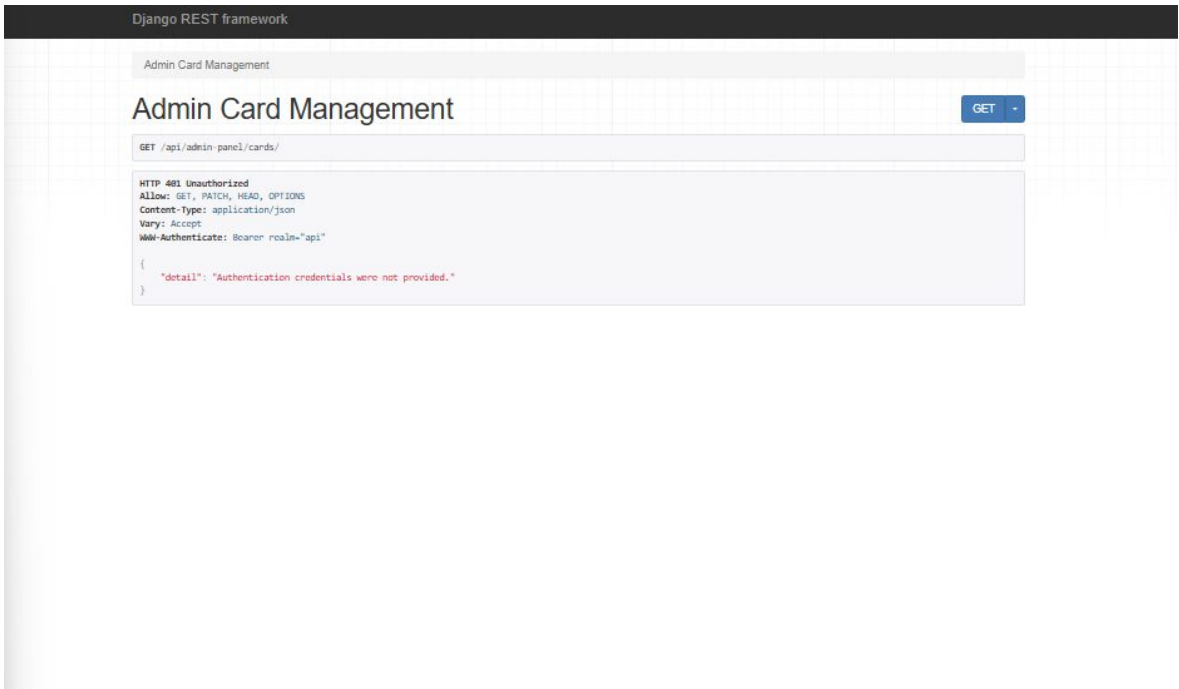
Role	Admin Dashboard	View Cards	Block / Unblock	Update Limit	Transaction Export
Admin	Yes	Yes	Yes	Yes	Yes
Support	Yes	Yes	Yes	Yes	No
Read-Only	Yes	Yes	No (403)	No (403)	No

Summary of Changes

- Created Admin, Support and Read-Only roles with permission classes applied to card management, transaction and analytics APIs.
- Unauthenticated requests return 401; users without the required role receive “You do not have permission to perform this action.”
- Added an audit log table recording the actor, action, target card, old/new value and timestamp for card block/unblock and credit-limit updates.
- Implemented fraud rules: multiple high-value transactions in a short time and rapid transactions from different locations/devices (3+ payments within 10 minutes).
- Added a fraud status field on transactions and a Fraud Detection Log (Open / Confirmed / False positive) with Confirm and Dismiss review actions.
- Email alerts are sent for flagged payments (“Suspicious Payment Alert”) and when an admin blocks a card (“Credit Card Blocked”).
- Built Card Usage Analytics APIs: monthly spending summary, category-wise expenses and credit utilization percentage, shown as line, bar and donut charts.
- Added advanced transaction search (date range, amount range, status, masked card number) with server-side pagination and sorting.
- Added API response-time and error logging and a System Health panel on the admin dashboard.
- Added analytics export to PDF and CSV, and an admin-only Transaction Export (CSV). Verified end to end using the DRF API, each role's dashboard and Gmail.

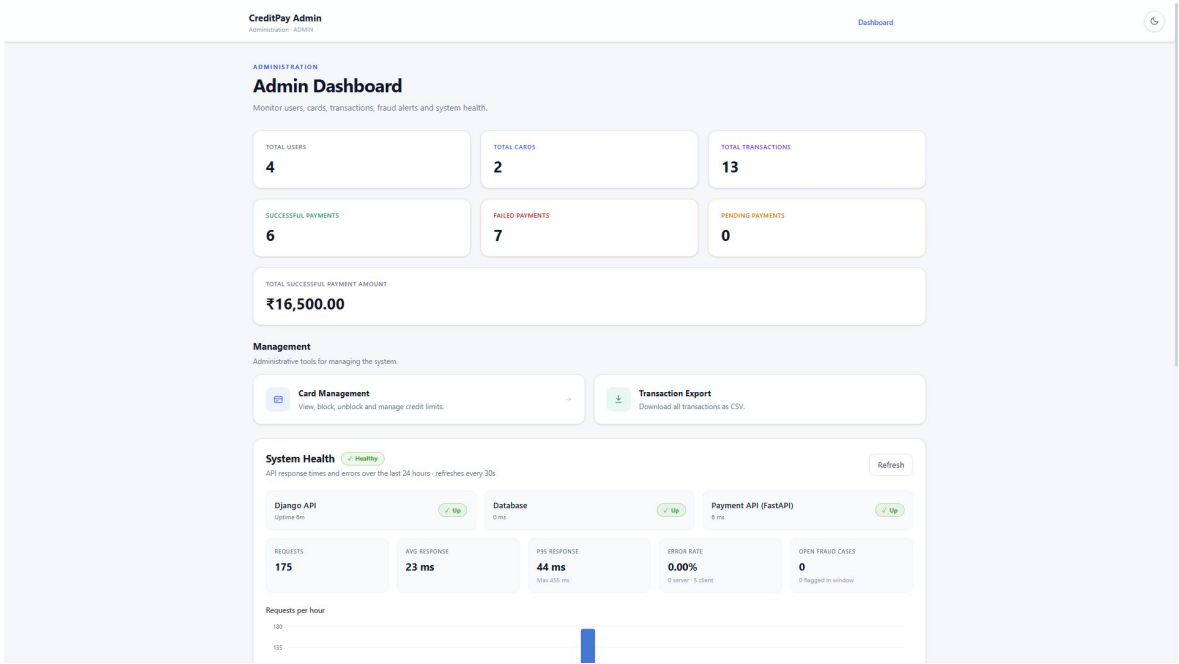
Implementation Screenshots

Role-Based Access Control — API Protection



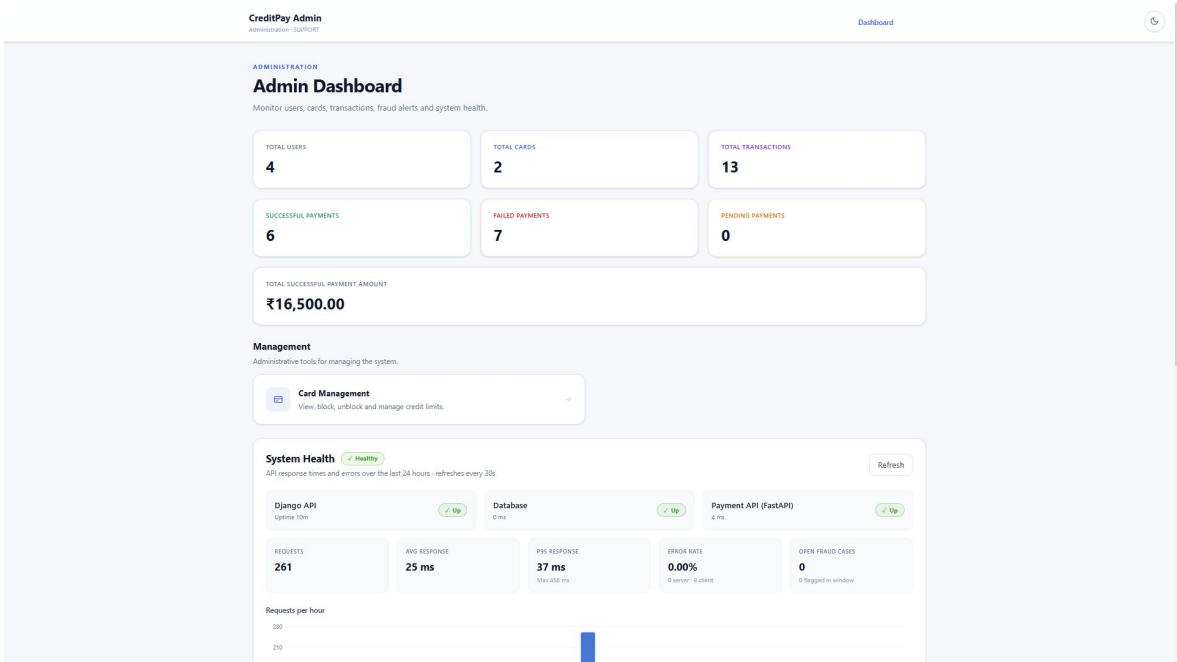
GET /api/admin-panel/cards/ returns 401 Unauthorized when no credentials are provided.

Admin Dashboard (Role: ADMIN)



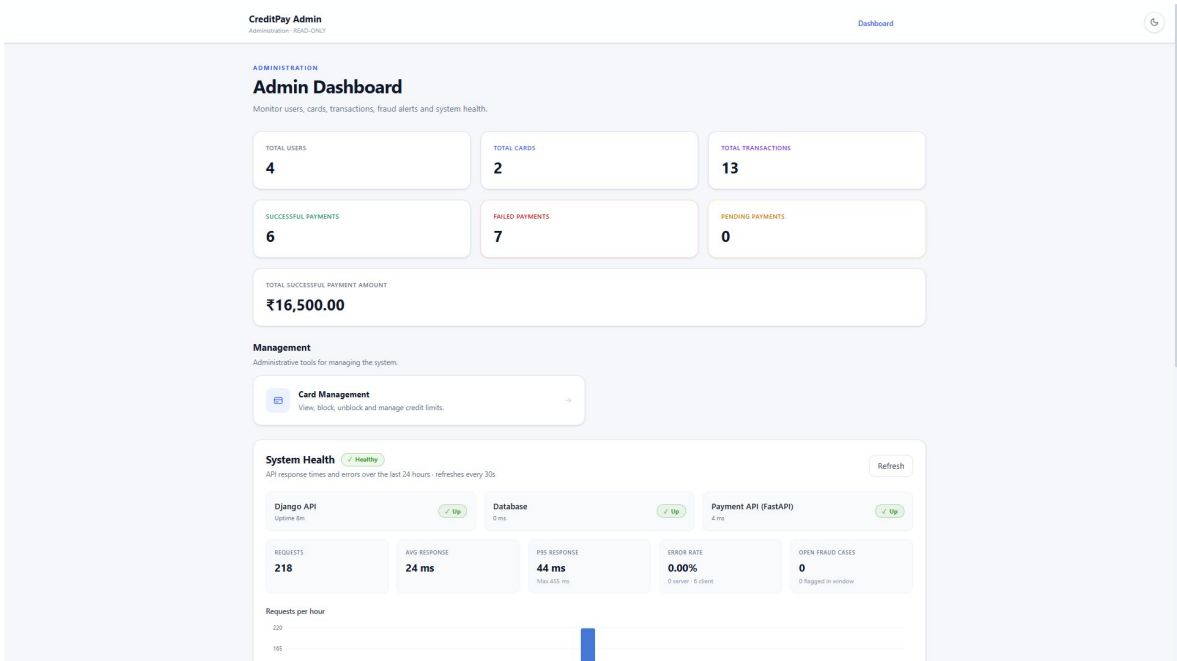
Admin dashboard with Card Management and Transaction Export available to the Admin role.

Support Dashboard (Role: SUPPORT)



Support role sees Card Management but not Transaction Export.

Read-Only Dashboard (Role: READ-ONLY)



Read-Only role can view the dashboard but has no export option.

Read-Only Role — Permission Checks

Card Management

CreditKey Admin

Admin Dashboard

ADMINISTRATION

Card Management

View all customer cards, manage credit limits, control card status and monitor card activity.

TOTAL CARDS

2

ACTIVE CARDS

2

BLOCKED CARDS

0

You do not have permission to perform this action.

*****4282

Card #6

ACTIVE

CARD HOLDER

Aprajit Roshan

ACCOUNT

testuser

CARD TYPE

DEBIT

CREDIT LIMIT

₹75,000.00

Block Card

New credit limit

Update Limit

Activity

*****1111

Card #7

ACTIVE

CARD HOLDER

Frontend Test User

ACCOUNT

testuser

CARD TYPE

CREDIT

CREDIT LIMIT

₹50,000.00

Block Card

New credit limit

Update Limit

Activity

Read-Only user blocked from card actions: “You do not have permission to perform this action.”

Card Management

CreditKey Admin

Admin Dashboard

ADMINISTRATION

Card Management

View all customer cards, manage credit limits, control card status and monitor card activity.

TOTAL CARDS

2

ACTIVE CARDS

2

BLOCKED CARDS

0

You do not have permission to perform this action.

*****4282

Card #6

ACTIVE

CARD HOLDER

Aprajit Roshan

ACCOUNT

testuser

CARD TYPE

DEBIT

CREDIT LIMIT

₹75,000.00

Block Card

100000

Update Limit

Activity

*****1111

Card #7

ACTIVE

CARD HOLDER

Frontend Test User

ACCOUNT

testuser

CARD TYPE

CREDIT

CREDIT LIMIT

₹50,000.00

Block Card

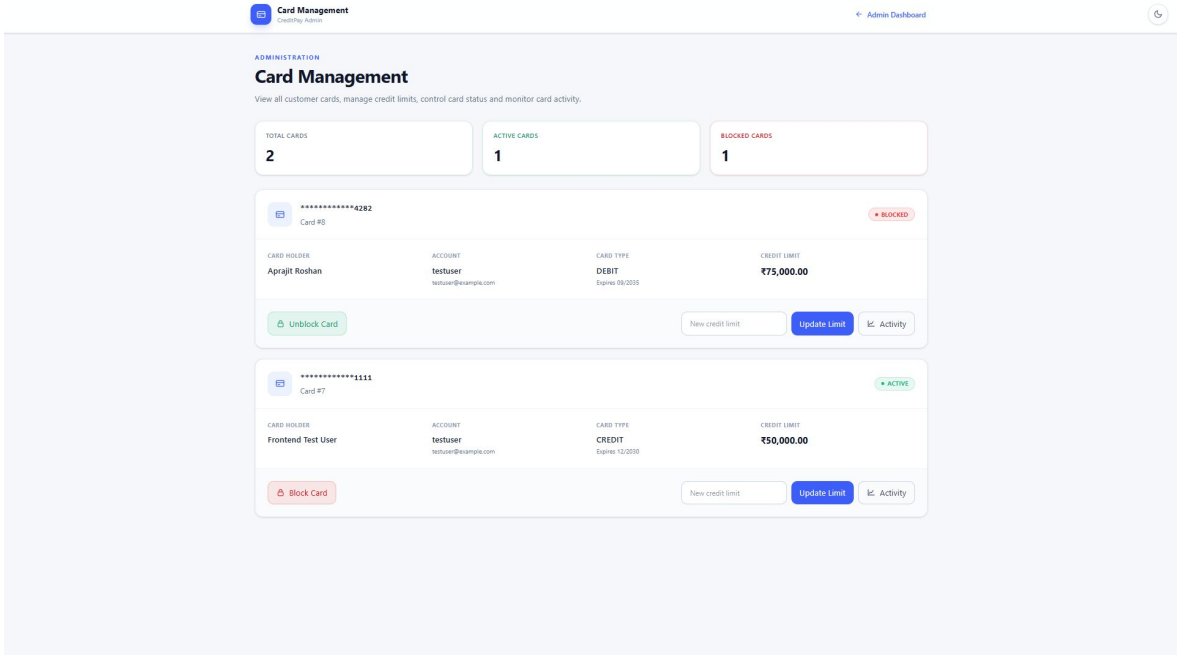
New credit limit

Update Limit

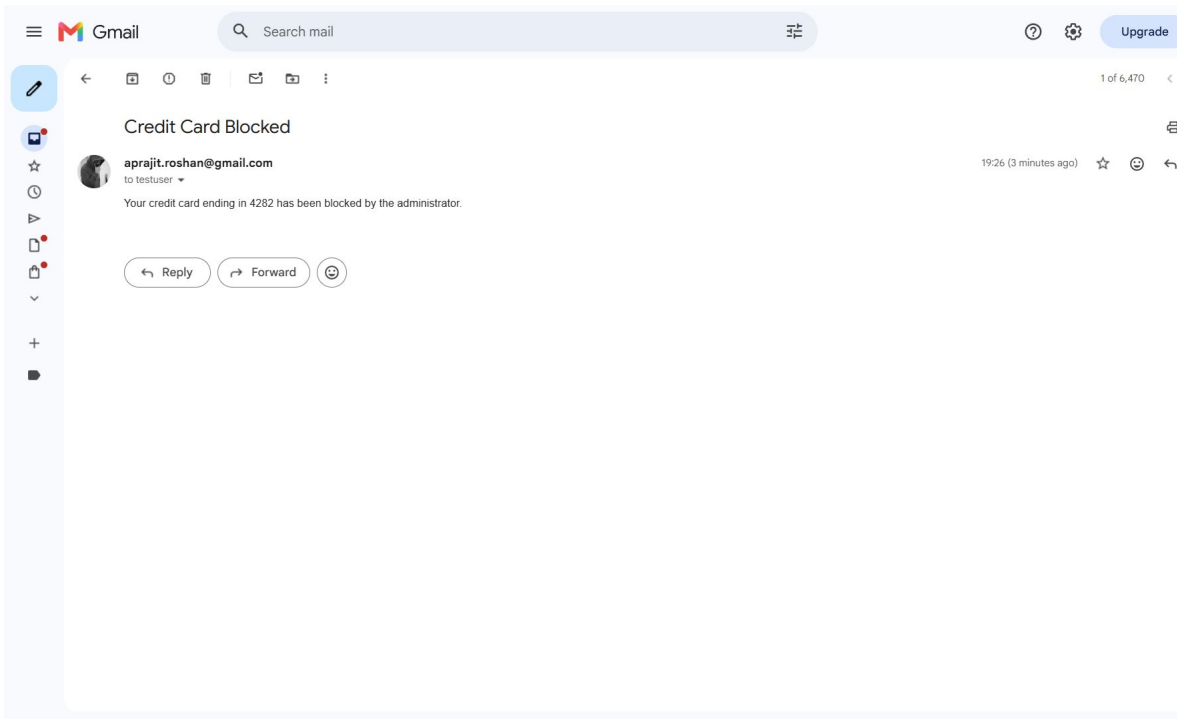
Activity

Credit limit update attempt (100000) also rejected for the Read-Only role.

Card Block — Admin Action and Email

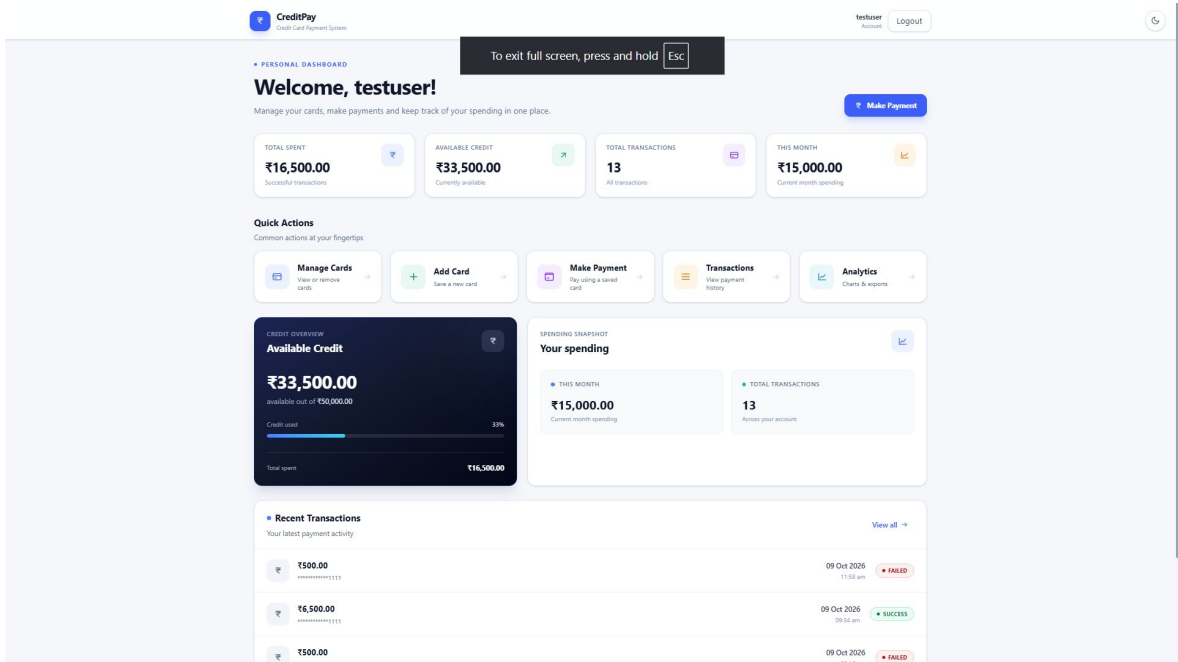


Admin blocks card ****4282: status changes to BLOCKED and the button changes to Unblock Card.



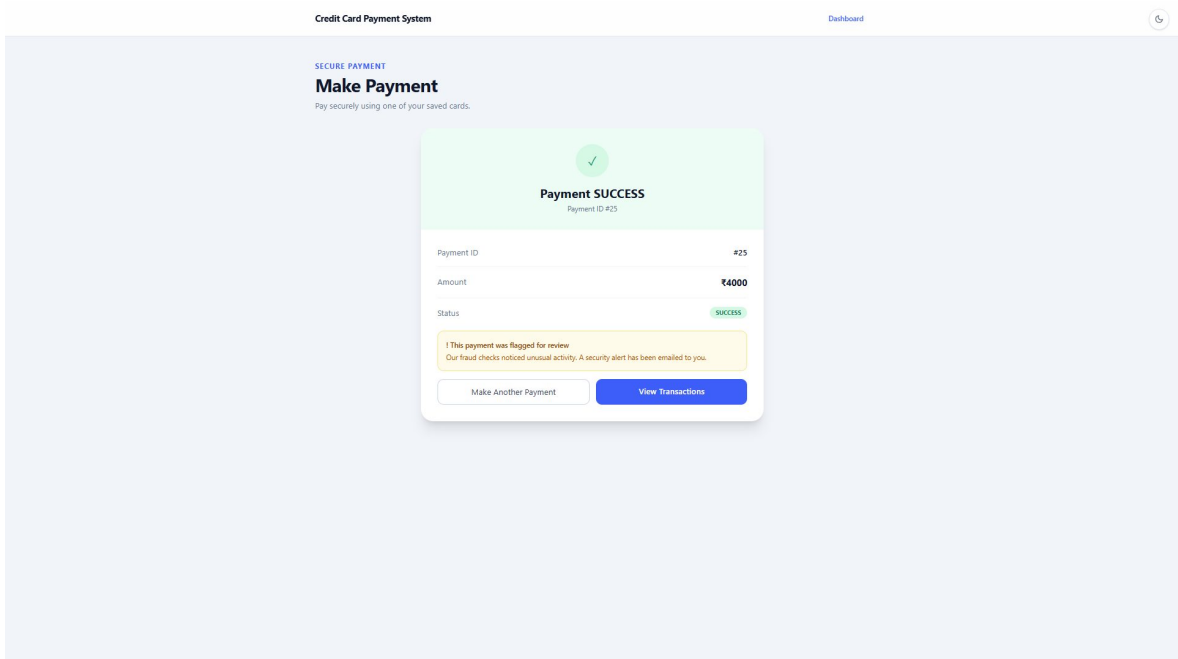
Customer receives the “Credit Card Blocked” email.

Customer Dashboard

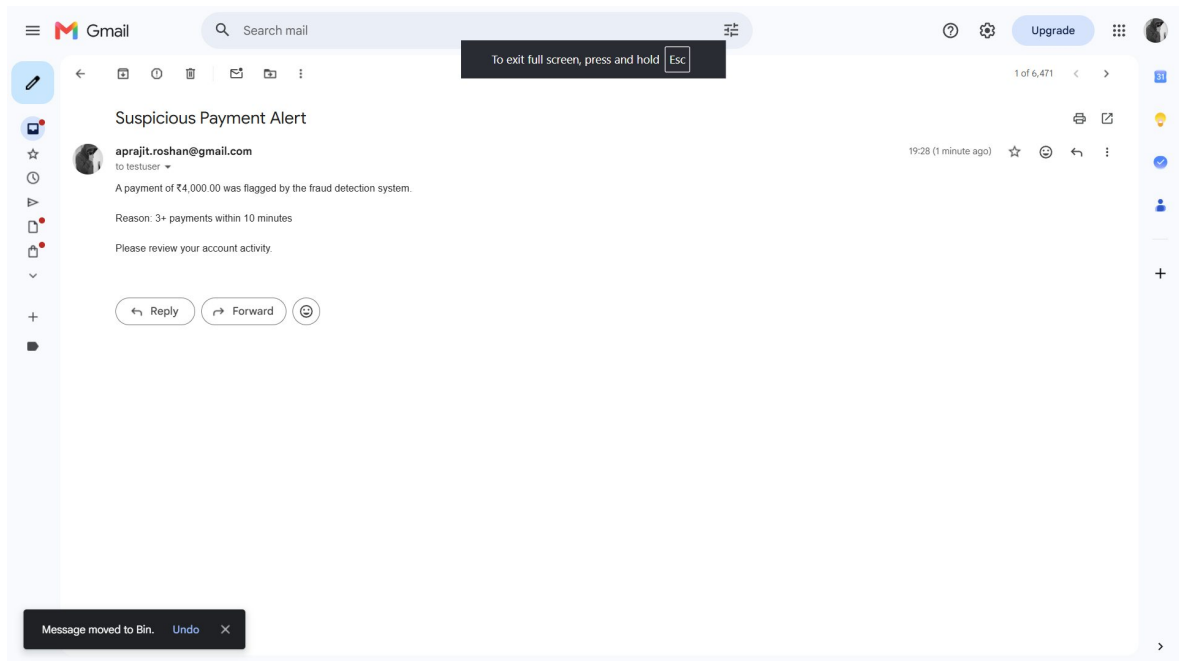


Customer dashboard showing total spent, available credit, credit used and recent transactions.

Fraud Detection — Flagged Payment and Alert

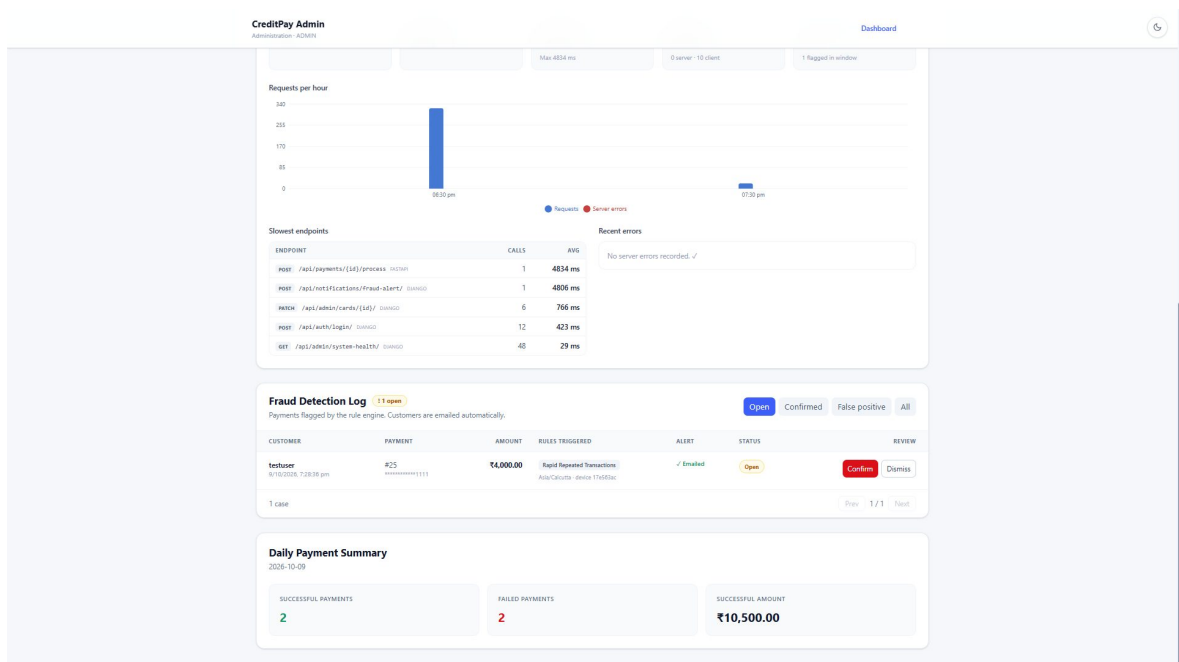


Payment #25 (Rs. 4000) succeeds but is flagged for review by the fraud checks.

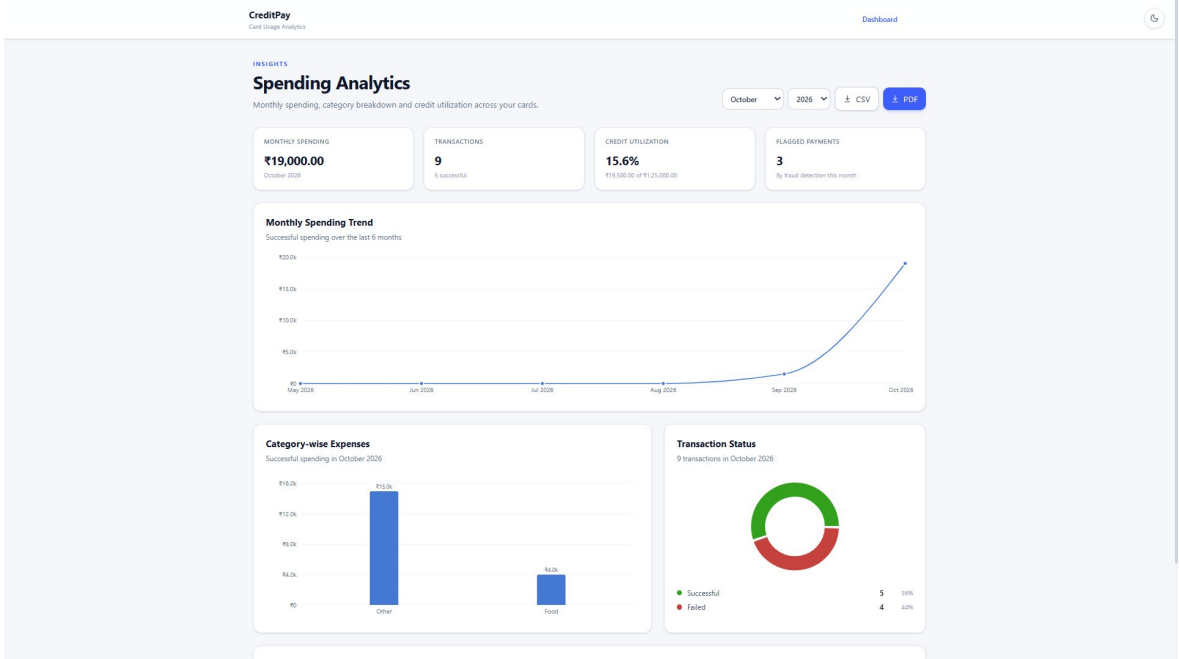


“Suspicious Payment Alert” email: 3+ payments within 10 minutes.

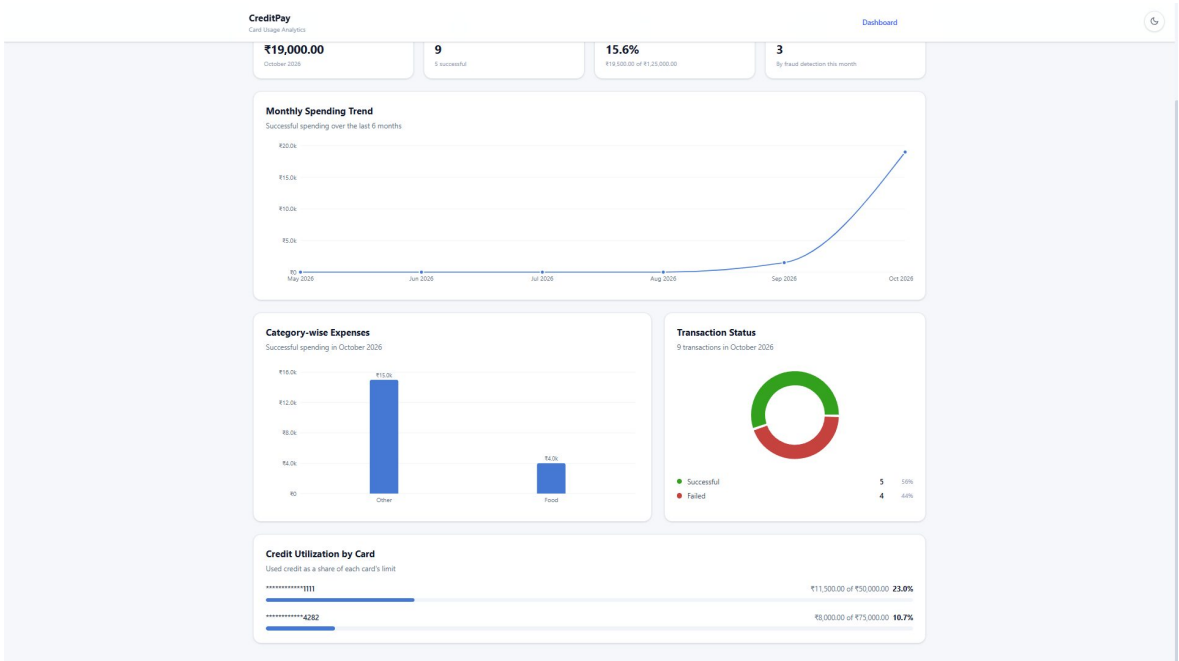
Fraud Detection Log



Spending Analytics

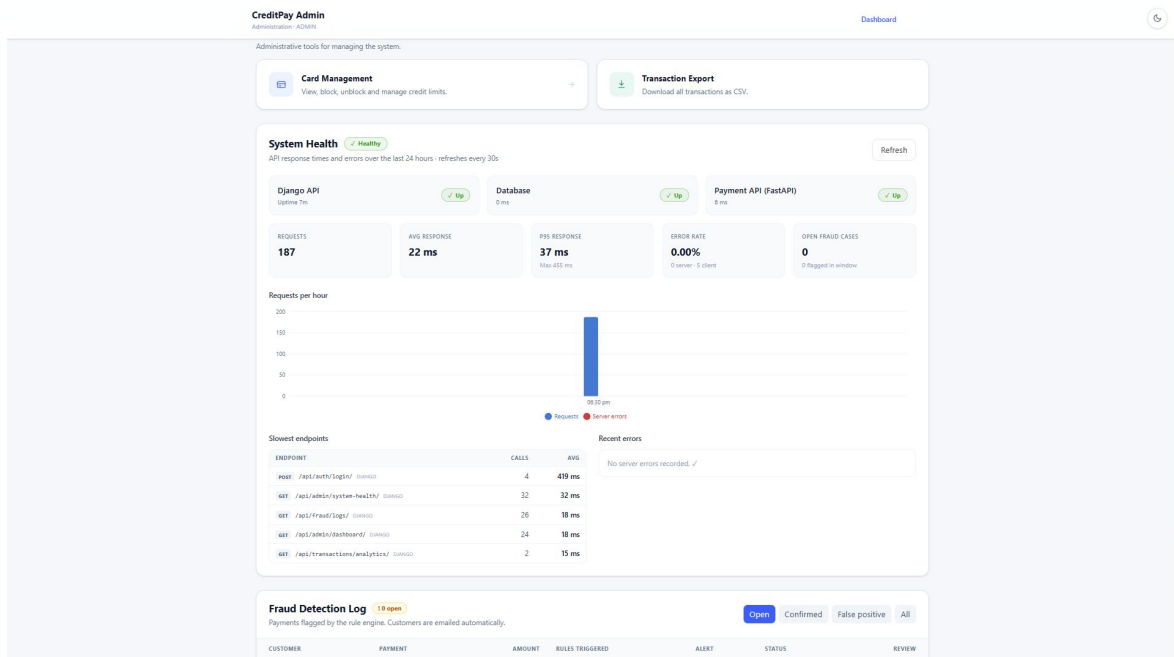


Monthly spending trend (line), category-wise expenses (bar) and transaction status (donut) with CSV / PDF export.

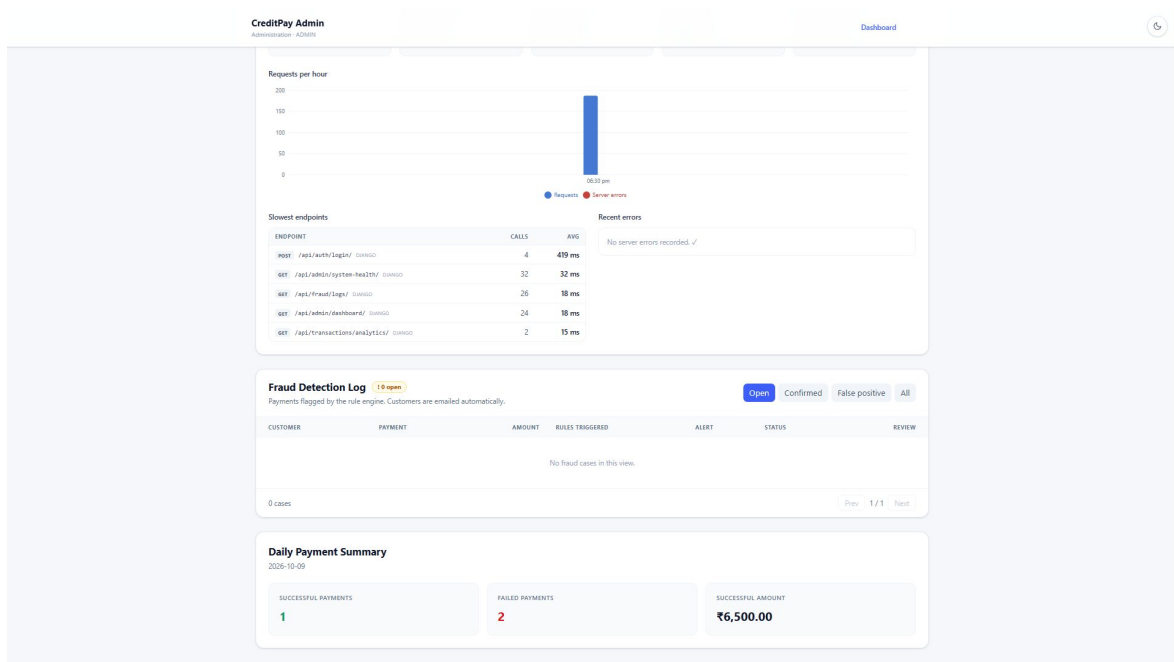


Credit utilization by card (****1111 at 23.0%, ****4282 at 10.7%).

System Health Monitoring



System Health: service status, response times, requests per hour, slowest endpoints and recent errors.



Fraud Detection Log (empty state) and Daily Payment Summary on the admin dashboard.

Conclusion: Role-based access control, fraud detection with email alerting, spending analytics with export, and system health monitoring have been implemented and verified end to end — each role sees and can do only what it is permitted to, suspicious payments are flagged, logged and emailed, and administrators can monitor system performance from a single dashboard.